

10 Inch Touch Screen Kit

Turnkey solution for local touchscreen control of Avi-on lighting control systems

High-Resolution

10" capacitive-touch screen with a 1920x1200 resolution with true-black display filter for exceptional clarity

Customizable Interface

Design the screen layout, configure the controller through the RS-232 Interface, and configure to control third-party AV devices

Flexible Mounting Options

Ultra-thin bezel for elegant mounting in landscape or portrait, or optional stand for countertop convenience

Control Unit & RS-232 Interface

The full range of Avi-on two-way control commands is supported via the RS-232 interface

The Avi-on® 10" Touch Screen Kit provides a turnkey solution for touch-screen control of Avi-on lighting control systems. **The touch screen is fully customizable** to support any desired layout and is available with both wall-mount and tabletop mount kits (optional).

The screen communicates through a system controller and the Avi-on RS-232 Serial Interface. **The full range of two-way Avi-on control commands is supported through the RS-232 interface.** Available in Black or White.

PRODUCT OVERVIEW

Installation

Mount the touch screen on the wall or use a tabletop mount kit. Mount the controller in a suitable location with reliable access to the Avi-on network (near lighting controllers or power packs). Connect an Ethernet cable between the touch screen and the controller using a router and wire the RS-232 Interface to the controller. Apply power to the controller and provide 12 VDC to the RS-232 Interface.

Note: A router is required to assign an IP to the touch screen.

Programming

Design the screen layout using the provided Windows-based software toolkit, and configure the controller interface to

communicate through the RS-232 Interface (standard serial configuration).

Program the control commands using the Avi-on Serial Control Protocol documentation. Some advanced control actions may require additional programming of the Avi-on network. Programming services for lighting control applications are available from Avi-on.

Applications

Ideal for conference rooms, control rooms, stadium command centers, and any location requiring a locally communicating, customizable touch-screen interface.

The touch screen can also be configured (by others) to control a wide range of third-party AV devices.

ORDERING INFORMATION

Part Number	Description	Application	Color
AVI-RTI-IST10-B	10" Touch Screen	Wall or Counter Control	Black
AVI-RTI-IST10-W	10" Touch Screen	Wall or Counter Control	White
AVI-KIT-RTI-XP3	Touch Screen Control Unit and RS232 Interface	Touch Screen Controller	-
AVI-RTI-IST10-FMK	Wall Mount Kit	Wall Mount	-
AVI-RTI-IST10-STAND	Table Mount Kit (Includes Power Supply, or use PoE)	Counter Mount	Silver

Project	Location/ Type
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SPECIFICATIONS

Touch Screen

Power:	(Power supply not included) +12 VDC, 2A, or PoE (802.3af)
Power Management:	Adjustable backlight / proximity sensor
Touchscreen Display:	High-brightness 24-bit color IPS LCD, WUXGA (1920 x 1200 pixels) with true-black display filter, Integrated high-resolution capacitive-touch screen
Display Size:	10" (254 mm) diagonal, 16:10 aspect ratio
Display Backlight:	LED backlit, 400 nit
Ethernet Port:	RJ45, 10/100/1000 Base-T
PoE Camera:	Color CMOS, 2 MP, 15–30 fps at 1920 x 1080
Microphone:	Built-in digital microphone array with mute button; hardware-based echo cancellation
Speakers:	Two (2), 8 Ω, 1 Watt, 100–20 kHz Communications
System Memory:	4 GB RAM, 64 GB ROM
Device Capability:	Total number of devices limited only by available memory
Macro Capability:	Total number of macros limited only by available memory
Operating Temperature:	+32°F to +122°F (0°C to +50°C)
Operating Humidity:	0.5% to 95% non-condensing
Mounting:	Wall-mount or aluminum table stand (available separately as a kit including power supply; not included)
Installation:	U.S. standard single or dual gang
Outer Bezel Dimension:	9.39" x 6.24" x 0.41"
(W x H x D)	(238.5 mm x 158.5 mm x 10.5 mm)
Wall Cut-Out Dimension:	4.765" x 4.5" (121.0 mm x 114.5 mm)
In Wall (Wall Box) Depth:	1.34" (34 mm); allow additional space for the RJ-45 connector
Weight:	14.1 oz (400 g)

Control Processor

Power Supply:	+12 VDC, 1A, or PoE (802.3af)
Multi-Purpose Output Port:	(Port 1) One custom 3.5 mm, 4-conductor jack. Designed for RTI modules as well as industry-standard IR emitters.
IR Output Ports:	(Ports 2 & 3) Two 3.5 mm jacks: Compatible with industry-standard IR emitters.
IR Output Drive:	100 mA maximum (per port, adjustable) Infrared
Frequency Transmission Range:	15 kHz – 460 kHz
RS-232 Port:	One, bi-directional
RJ45 Ethernet Port:	One 10/100Base-T with Power-over-Ethernet (Class 3)
Relays:	Two, 5 Amps @ 30 VDC or 12 V / 100 mA
Trigger Sense Inputs:	Two, 3–24 VDC, Off/Low voltage condition must not exceed 1 VDC
USB Ports:	One
Mounting:	Wall-mount or free-standing
Operating Temperature:	+32°F to +122°F (0°C to +50°C)
Operating Humidity:	0.5% to 95% non-condensing
Dimensions (W x H x D):	7.0" x 3.0" x 1.0" (177 mm x 76 mm x 25 mm)
Weight:	6.4 oz (100 g)

RS-232 Serial Interface

Input Voltage:	12–24 VDC
Power Consumption:	17 mA
CC Input:	12–24 VDC non-powered contact closure detection (momentary)
Direct Connect Input:	Compatible with all Avi-on Direct Connect™ Motion/Daylight Sensors
Size:	2.30 in x 1.43 in x 0.75 in (58.2 mm x 36.4 mm x 19 mm)
Mounting:	Removable mounting tabs
Weight:	0.45 oz (16 g)
Terminal Blocks:	22–16 AWG wires
Operating Temperature:	–22°F to +158°F (–30°C to +70°C)
Storage Temperature:	–40°F to +185°F (–40°C to +85°C)
Humidity Rating:	95% non-condensing
Radio Frequency:	2.4 GHz
Wireless Standard:	BLE 5.3 with Mesh
Point-to-Point Range*:	80 ft with obstructions; 350 ft unobstructed
Security:	AES 128-bit encryption for device-to-device communication AES 256-bit encryption for device-to-cloud communication
Warranty:	5 years; 10 years optional
Regulatory:	FCC ID: 2AFZI-AVIBG21 IC ID: 20544-AVIBG21 BQB: DID: D063032 CE: EN IEC 61000-6-1:2019, EN IEC 61000-6-3:2021

*When communicating through the mesh, range is essentially unlimited (5000ft+)

Wall Mount

Dimensions:	12.80" x 9.45" x 1.73" (W x H x D) 325.1 mm x 240 mm x 44 mm
Shipping Dimensions:	13.77" x 10.6" x 2.16" (W x H x D) 350 mm x 270 mm x 55 mm
Weight:	0.7 kg (1.54 lbs)
Material:	Moisture-resistant MDF (compliant with TSCA Title VI)
Installation:	To be installed in ½" drywall

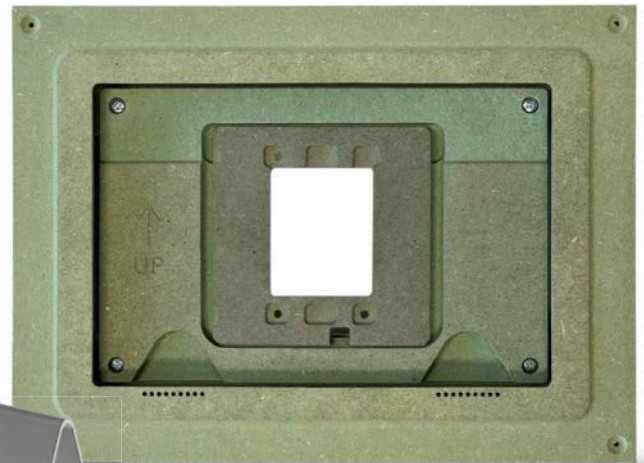
Table Mount

Color:	Silver anodized aluminum, black rubber feet
Dimensions (H x L x W):	5.25" (133.4 mm) x 5.25" (133.4 mm)
Front height:	5.27" (144.8 mm)
Rear height:	2.75" (69.9 mm)
Weight:	7.05 oz (200 g)
Included in the Box:	IST stand, mounting screws (4 + 2 spare), 12 VDC / 2A power supply

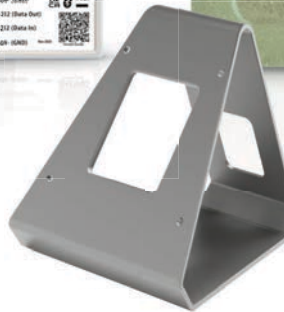
ADDITIONAL IMAGES



**Integrated Central Processor +
Serial Interface**
AVI-KIT-RTI-XP3

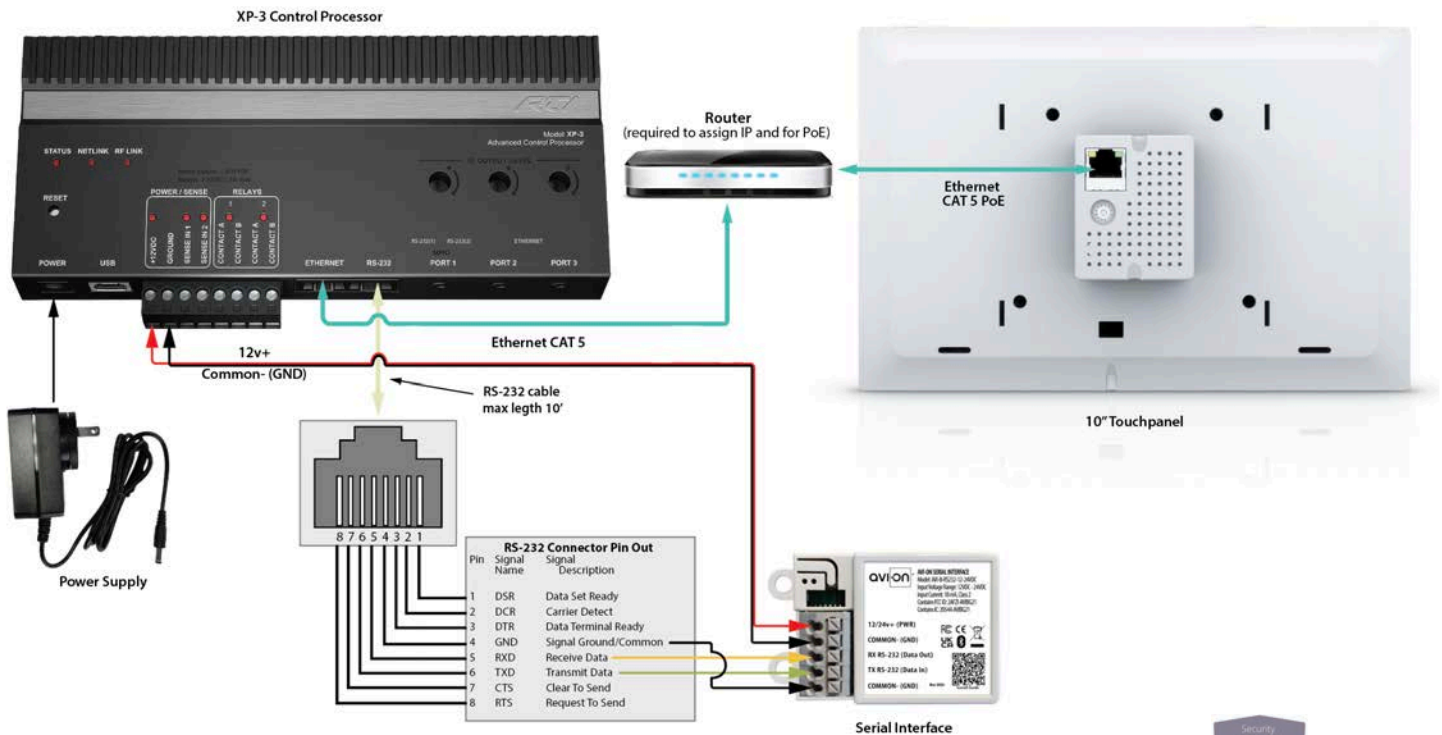


Flush Mount Kit
AVI-RTI-IST10-FMK



Stand Kit
AVI-RTI-IST10-STAND

WIRING



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